

COMPUTER SOFTWARE CAT 2 MARKING SCHEME

1. Importance of software testing in ensuring software functionality. (4 Marks)

- Ensures software works as intended according to requirements – 1 Mark
- Identifies bugs and errors before deployment – 1 Mark
- Improves software quality and reliability – 1 Mark
- Ensures user satisfaction and system stability – 1 Mark

2. FOUR reasons why software testing is important in system development. (4 Marks)

Any FOUR:

- Detects defects early
- Reduces development cost
- Improves software performance
- Ensures security of the system
- Enhances user satisfaction
- Ensures compliance with requirements

(1 Mark each)

3. Functional vs Non-functional testing. (4 Marks)

- Functional testing checks what the system does (features/functions) – 2 Marks
- Non-functional testing checks how the system performs (speed, usability, reliability, performance) – 2 Marks

OR suitable comparison

4. THREE software testing techniques used in verifying software functionality. (3 Marks)

Any THREE:

- Black box testing
- White box testing
- Gray box testing

- Unit testing
- Integration testing

(1 Mark each)

5. TWO advantages of automated testing over manual testing. (3 Marks)

Any TWO:

- Faster execution of tests – 1½ Marks
- More accurate and reduces human error – 1½ Marks
- Can run repeated tests easily
- Saves time and cost in long-term

6(a). Test Complete (2 Marks)

- Automated testing tool used to test desktop, web, and mobile applications – 2 Marks

6(b). Selenium (2 Marks)

- Open-source tool used for automating web application testing – 2 Marks

6(c). Postman (2 Marks)

- Tool used for testing APIs by sending requests and analyzing responses – 2 Marks

7. What is Postman and its use in software testing. (3 Marks)

- Postman is an API testing tool – 1 Mark
- Used to send HTTP requests (GET, POST, PUT, DELETE) – 1 Mark
- Used to test, debug, and validate APIs – 1 Mark

8. Step-by-step use of Postman to test an API. (5 Marks)

Any correct steps:

1. Open Postman application – 1 Mark
2. Create a new request – 1 Mark
3. Select HTTP method (GET, POST, etc.) – 1 Mark
4. Enter API URL – 1 Mark
5. Click Send and analyze response – 1 Mark

9. FOUR HTTP methods used in Postman. (4 Marks)

Any FOUR:

- GET – Retrieve data – 1 Mark
- POST – Send/create data – 1 Mark
- PUT – Update data – 1 Mark
- DELETE – Remove data – 1 Mark

10. TWO benefits of API testing using Postman. (3 Marks)

Any TWO:

- Ensures API functionality works correctly – 1½ Marks
- Detects errors early in development – 1½ Marks
- Improves reliability of services
- Supports automation of tests

11. Steps involved in testing a new software application before deployment. (4 Marks)

Any FOUR:

1. Requirement analysis – 1 Mark
2. Test planning – 1 Mark
3. Test case development – 1 Mark
4. Test execution – 1 Mark
5. Reporting and debugging (acceptable alternative)

12. THREE common errors detected during software functionality testing. (3 Marks)

Any THREE:

- Syntax errors
- Runtime errors
- Logic errors
- Interface errors
- Performance errors

(1 Mark each)

13. TWO challenges faced during software testing and solutions. (3 Marks)

Challenge 1 + solution – 1½ Marks

Challenge 2 + solution – 1½ Marks

Examples:

- Lack of test data → create proper test datasets
- Time constraints → automate testing
- Incomplete requirements → improve documentation
- Environment issues → set up proper test environment

14. Role of regression testing in maintaining software functionality. (3 Marks)

- Ensures new changes do not break existing functionality – 1 Mark
- Confirms previously working features still work – 1 Mark
- Used after updates, bug fixes, or enhancements – 1 Mark

15. Role of test cases in software functionality testing. (3 Marks)

- Defines steps to test software – 1 Mark
- Specifies expected results – 1 Mark
- Helps ensure consistent and repeatable testing – 1 Mark

TOTAL: 50 MARKS